

Bangladesh University of Engineering and Technology (BUET)

Simulation Report

FF-AODV: Energy and Velocity Aware Routing in ns-3.39

Course: CSE 322 – Computer Networks Sessional
Topic: Routing protocol modification and performance evaluation
Simulator: ns-3.39 (C++)
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Abstract

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This report presents a complete ns-3.39 based simulation study of a modified AODV protocol named FF-AODV (Fitness Function AODV). The assignment-required static topologies are covered using (i) IEEE 802.11b ad-hoc and (ii) an IEEE 802.15.4-like emulation setup. A bonus mobile FANET scenario is additionally evaluated using IEEE 802.11a with Gauss-Markov mobility. The protocol is implemented in two phases: Phase 1 (energy + hop aware) and Phase 2 (energy + hop + velocity aware). The report includes topology descriptions, parameter variations, exact simulator-level changes, results with graphs, summary findings, and reflection/discussion. Results show expected degradation under larger areas and higher offered load, while the Phase 2 mobility-aware extension provides only modest changes over Phase 1 in the currently available speed-sweep dataset, indicating that higher statistical repetition and broader stress conditions are needed for stronger conclusions.

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1. Introduction

Standard AODV prioritizes route discovery and freshness but fundamentally prefers routes by sequence number and hop count related logic. In mobile and energy constrained wireless settings, this can be suboptimal because:

1. low-energy relays can be repeatedly overused,
2. high-speed relays can destabilize routes,
3. shortest path alone does not capture route robustness.

To address this, FF-AODV introduces a fitness score that combines residual energy, path efficiency, and in Phase 2, velocity stability. The main goals of this report are:

1. present all assignment-required simulation topologies and parameter sweeps,
2. document simulator code modifications in detail,
3. analyze result trends using generated graphs and CSV values,
4. summarize findings and discuss limitations and future improvements.

2. Network Topologies Under Simulation

2.1 Topology A: Wireless 802.11 Static

This topology uses the program `scratch/sim-802-11-static.cc`. Key characteristics are:

- Standard: IEEE 802.11b ad-hoc (DSSS 1 Mbps fixed rate).
- Mobility: static nodes (`ConstantPositionMobilityModel`).
- Placement: random uniform in square area of side $\text{areaFactor} \times \text{txRange}$.
- Traffic: UDP OnOff, packet size 512 bytes.
- Energy: `BasicEnergySource` (100 J) + `WifiRadioEnergyModel`.
- Routing settings (Phase 1 mode): $\alpha = 0.6$, $\beta = 0.4$, $\gamma = 0.0$.

2.2 Topology B: 802.15.4-like Static Emulation

This topology uses `scratch/sim-802-15-4-static.cc`. It is intentionally configured to emulate low-rate, short-range behavior:

- PHY/MAC base: 802.11b ad-hoc with reduced Tx power (9 dBm).
- Emulation intent: short range and small payload behavior similar to 802.15.4.
- Packet size: 80 bytes (close to practical 802.15.4 payload limits).
- Area base: `txRange=50 m` (smaller than 802.11 setup).
- Mobility: static.
- Routing settings: $\alpha = 0.6$, $\beta = 0.4$, $\gamma = 0.0$.

Important implementation note: true IPv4 + AODV over LR-WPAN + 6LoWPAN is stated in code comments as problematic for this setup in ns-3.39, hence this controlled emulation was used.

2.3 Topology C (Bonus): FANET Mobile Scenario

Two scripts are used for comparison:

- `scratch/sim-fanet-phase1-bonus.cc` (Phase 1 baseline),
- `scratch/sim-fanet-bonus.cc` (Phase 2 velocity-aware).

Common FANET settings:

- Standard: 802.11a ad-hoc at 6 Mbps.
- Mobility: `GaussMarkovMobilityModel` in 3D bounds $500 \times 500 \times 100$.
- Energy: 100 J per node with WiFi radio model.
- Traffic: UDP OnOff, packet size 512 bytes.
- Variable of interest: node speed (5, 10, 15, 20, 25 m/s).

3. Parameters Under Variation

3.1 Assignment-Aligned Parameter Sweeps

The mandatory static study varied one parameter at a time while keeping others fixed.

Table 1: Parameter values used in the sweeps

Parameter	Values
Number of nodes	20, 40, 60, 80, 100
Number of flows	10, 20, 30, 40, 50
Packets per second	100, 200, 300, 400, 500
Coverage area factor (static)	1, 2, 3, 4, 5
Node speed (mobile bonus)	5, 10, 15, 20, 25 m/s

3.2 Fixed Baselines per Sweep

From the automation scripts:

- Static defaults: $nNodes = 40$, $nFlows = 10$, $pktPerSec = 100$, $areaFactor = 1$.
- FANET phase-comparison defaults: $nNodes = 60$, $nFlows = 15$, $pktPerSec = 20$, multi-seed runs.

3.3 Measured Metrics

All required metrics are captured:

1. Network throughput (Mbps)
2. Average end-to-end delay (ms)
3. Packet delivery ratio (PDR, %)
4. Packet drop ratio (PDR, %)
5. Total energy consumption (J)

4. Modifications Made in the Simulator

4.1 Fitness Function Design

Phase 1 (Energy + Hop)

$$F_{p1} = \alpha \cdot \frac{E_{res}}{E_{init}} + \beta \cdot \frac{1}{h} \quad (1)$$

where $\alpha = 0.6$, $\beta = 0.4$ in static experiments.

Phase 2 (Energy + Hop + Velocity)

$$F_{p2} = \alpha \cdot \frac{E_{res}}{E_{init}} + \beta \cdot \frac{1}{h} + \gamma \cdot \frac{V_{max} - V}{V_{max}} \quad (2)$$

with Phase 2 defaults: $\alpha = 0.5$, $\beta = 0.3$, $\gamma = 0.2$, $V_{max} = 50$ m/s.

The path fitness is propagated using a weakest-link principle:

$$F_{path} = \min(F_{incoming}, F_{local}) \quad (3)$$

and the path velocity field tracks worst-case mobility:

$$V_{path} = \max(V_{incoming}, V_{local}) \quad (4)$$

4.2 AODV Data Structure and Packet-Level Changes

The following simulator-level changes were made inside `src/aodv`:

- Added path fitness and velocity fields to RREQ and RREP headers.
- Extended (de)serialization for both fields:
 - RREQ size became 39 bytes (23 + 8 + 8).
 - RREP size became 35 bytes (19 + 8 + 8).
- Added route-entry fitness field in routing table entries.

4.3 Routing Decision Logic Changes

Core behavioral changes in routing protocol implementation include:

1. Reverse-route updates in RREQ processing use fitness comparison.

2. Duplicate RREQ handling is relaxed: duplicates are accepted only if fitness improves.
3. RREP path fitness/velocity are seeded and propagated so destination-side contribution is preserved.
4. When sequence numbers tie in RREP update logic, route preference is based on better fitness (instead of hop-only preference).

4.4 Build Integration

The AODV module build now links with energy library support (`$libenergy`) in `src/aodv/CMakeLists.txt` to enable direct residual-energy access in routing logic.

4.5 Files Modified (Summary)

File	Main Modification
<code>src/aodv/model/aodv-packet.h</code>	Added fitness and velocity fields with getters/setters in RREQ and RREP headers.
<code>src/aodv/model/aodv-packet.c</code>	Serialization/deserialization updates for new fields; updated header sizes and print/equality behavior.
<code>src/aodv/model/aodv-rtable.h</code>	Added route fitness member and accessor methods.
<code>src/aodv/model/aodv-rtable.c</code>	Initialized route fitness default to 0.0.
<code>src/aodv/model/aodv-routing-protocol.c</code>	Added velocity-related parameters and updated fitness function signature.
<code>src/aodv/model/aodv-routing-protocol.h</code>	Implemented fitness calculation, energy and mobility extraction, RREQ/RREP propagation and update rules.
<code>src/aodv/CMakeLists.txt</code>	Linked energy library dependency.

5. Results With Graphs

5.1 Overview Graphs (Mandatory Static Study)

The following all-metric dashboards aggregate throughput, delay, PDR, drop ratio, and energy for each variation axis.

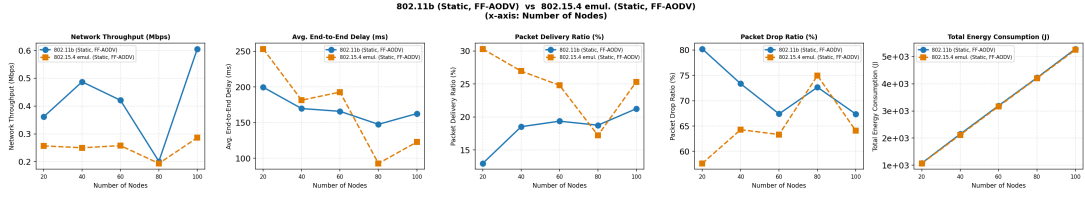


Figure 1: 802.11 vs 802.15.4-like emulation: all metrics while varying number of nodes

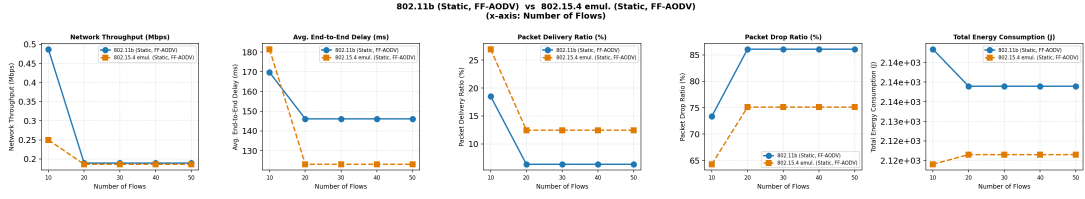


Figure 2: 802.11 vs 802.15.4-like emulation: all metrics while varying number of flows

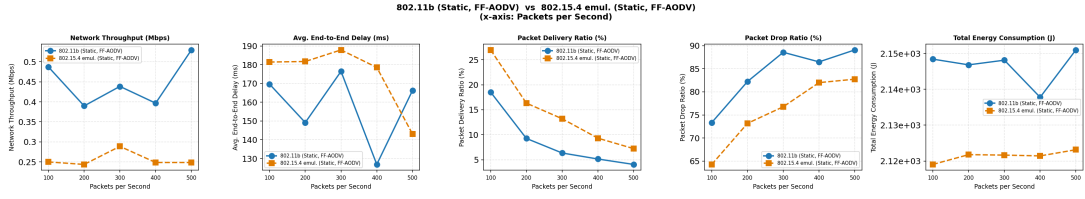


Figure 3: 802.11 vs 802.15.4-like emulation: all metrics while varying packet rate

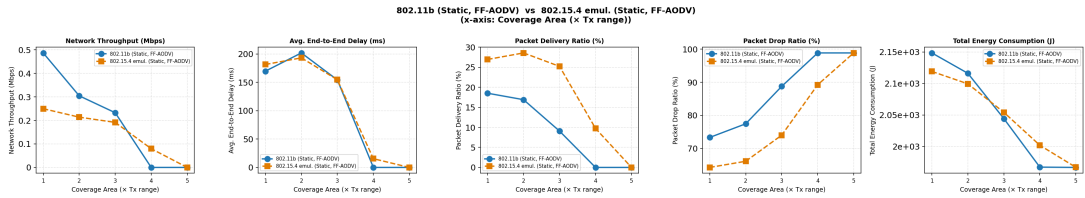


Figure 4: 802.11 vs 802.15.4-like emulation: all metrics while varying coverage area

5.2 Numerical Results: Static Topologies

A. Varying Number of Nodes

Table 3: Node sweep results (static, area factor=1, flows=10, pps=100)

Nodes	T11	T154	D11	D154	PDR11	PDR154	Drop11	Drop154
20	0.361	0.257	199.618	253.406	12.955	30.321	80.207	57.623
40	0.487	0.250	169.697	181.450	18.528	26.959	73.344	64.285
60	0.421	0.258	165.836	192.851	19.361	24.804	67.394	63.322
80	0.201	0.193	147.577	92.692	18.755	17.212	72.652	74.957
100	0.605	0.286	162.614	122.674	21.254	25.343	67.354	64.058

Observation: Throughput is generally higher in 802.11. Both systems show non-monotonic behavior due random placement and single-run per point in static sweeps. Energy consumption (not shown in this compact table) grows with node count in both systems, reaching about 5281 J (802.11) and 5246 J (802.15.4-like) at 100 nodes.

B. Varying Number of Flows

Table 4: Flow sweep results (static, nodes=40, area factor=1, pps=100)

Flows	T11	T154	D11	D154	PDR11	PDR154	Drop11	Drop154
10	0.487	0.250	169.697	181.450	18.528	26.959	73.344	64.285
20	0.189	0.186	146.149	123.123	6.346	12.453	86.087	75.114
30	0.189	0.186	146.149	123.123	6.346	12.453	86.087	75.114
40	0.189	0.186	146.149	123.123	6.346	12.453	86.087	75.114
50	0.189	0.186	146.149	123.123	6.346	12.453	86.087	75.114

Observation: The flat values from 20 to 50 flows are a direct artifact of implementation capping: $\text{actualFlows} = \min(n\text{Flows}, n\text{Nodes}/2)$. With $n\text{Nodes} = 40$, the effective maximum is 20 flows, so requested values above 20 do not create additional offered load.

C. Varying Packets per Second

Table 5: Packet-rate sweep results (static, nodes=40, flows=10, area factor=1)

PPS	T11	T154	D11	D154	PDR11	PDR154	Drop11	Drop154
100	0.487	0.250	169.697	181.450	18.528	26.959	73.344	64.285
200	0.390	0.244	149.182	181.747	9.266	16.321	82.233	73.176
300	0.439	0.289	176.482	187.912	6.358	13.209	88.558	76.769
400	0.397	0.248	126.850	178.586	5.156	9.321	86.485	81.994
500	0.530	0.248	166.312	143.060	4.072	7.231	89.071	82.738

Observation: Both systems suffer PDR reduction and drop-ratio increase with rising offered packet rate. 802.11 maintains higher throughput envelope, while 802.15.4-like stays lower and more constrained.

D. Varying Coverage Area

Table 6: Area sweep results (static, nodes=40, flows=10, pps=100)

AreaF	T11	T154	D11	D154	PDR11	PDR154	Drop11	Drop154
1	0.487	0.250	169.697	181.450	18.528	26.959	73.344	64.285
2	0.305	0.214	201.835	193.172	16.914	28.585	77.464	66.145
3	0.233	0.192	154.255	154.538	9.076	25.230	88.769	74.021
4	0.000	0.080	0.000	15.424	0.000	9.738	98.877	89.252
5	0.000	0.000	0.000	0.000	0.000	0.000	98.877	98.877

Observation: Area expansion strongly harms connectivity. At larger area factors, both networks approach near-disconnection behavior (very low throughput and near-total packet loss).

5.3 Bonus Results: FANET Phase 1 vs Phase 2

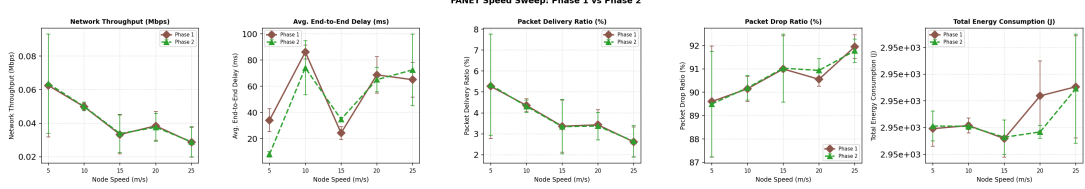


Figure 5: FANET comparison dashboard: Phase 1 (energy+hop) vs Phase 2 (energy+hop+velocity)

Table 7: Mean FANET metrics by speed (current available runs)

Speed (m/s)	T(P1) Mbps	T(P2) Mbps	Delay(P1) ms	Delay(P2) ms	PDR(P1) %	PDR(P2) %
5	0.062442	0.063529	34.074	8.076	5.271	5.342
10	0.049755	0.049951	86.100	74.040	4.361	4.315
15	0.033288	0.033839	24.264	34.786	3.365	3.345
20	0.038421	0.037574	68.691	65.022	3.431	3.372
25	0.028615	0.028945	64.974	72.468	2.616	2.641

Observation: In the present dataset, Phase 2 and Phase 1 are very close in throughput and PDR across speeds, with mixed delay differences. This indicates modest net effect under the chosen stress level and sample count, not a clear universal gain or loss.

6. Summary Findings

- Topology behavior:** Both static topologies degrade with higher area and traffic stress; 802.11 generally provides higher throughput than the 802.15.4-like emulation.
- Connectivity sensitivity:** Coverage-area increase is the strongest stress factor; both topologies approach near-total drop around the highest area factors.
- Protocol modification correctness:** FF-AODV fitness propagation is integrated consistently into RREQ/RREP logic, route-table storage, and tie-break route updates.
- Bonus mobility comparison:** Phase 2 velocity-aware routing currently shows small, mixed differences versus Phase 1 in available mean results.
- Energy trend:** Total consumed energy increases with network size/load as expected; mobility-phase comparison shows only minor energy differences between phases.

7. Reflection and Discussion

7.1 What Worked Well

- The simulator modifications are deep (packet format, routing table, route-update logic) and therefore academically meaningful.
- The full automation pipeline (simulation + plotting) improves reproducibility and minimizes manual reporting error.
- Weakest-link path fitness and worst-case path velocity are conceptually aligned with stability-aware routing.

7.2 Important Limitations in Current Results

1. **Flow-sweep capping artifact:** in static scripts, effective flow count is capped at $nNodes/2$; therefore requested flows above 20 (for $nNodes = 40$) do not actually increase offered load.
2. **Single-run static points:** non-monotonic curves are expected when each point has only one random realization.
3. **802.15.4 representation:** this is an emulation through tuned WiFi behavior, not a full LR-WPAN + IPv6 routing stack experiment.
4. **FANET sample depth:** current available phase-comparison CSV has limited run count, so confidence intervals are not yet tight.

7.3 Interpretation of Phase 2 Outcome

The velocity-aware term does not guarantee universal improvement in every metric at every speed. This is expected in MANET/FANET studies where route quality, contention, and mobility interact nonlinearly. A robust claim requires more seeds and scenario diversity.

7.4 Recommended Next Steps

1. Remove or parameterize the flow capping effect for true 10–50 flow stress tests.
2. Run 5–10 seeds for all static points and report mean with variance bars.
3. Add confidence intervals for FANET Phase 1 vs Phase 2.

4. Evaluate additional mobility models and larger offered loads.

8. Reproducibility Commands

All results and graphs in this project are generated from scripts in `ns-allinone-3.39/ns-3.39`.

```
# From ns-allinone-3.39/ns-3.39
./run_all_simulations.sh
./run_fanet_phase_comparison.sh
python3 plot_graphs.py
python3 plot_fanet_phase_comparison.py
```

9. Conclusion

This work successfully implements and evaluates FF-AODV in ns-3.39 with assignment-compliant static experiments and a mobility-aware FANET extension. The report demonstrates end-to-end protocol engineering: metric design, packet/routing state extensions, routing-decision rewrites, experiment automation, and graph-backed analysis. The most impactful practical insight is that methodological details (effective load, random-seed count, and topology realism) strongly influence interpretation; therefore, simulator modifications and experimental design must be validated together for defensible conclusions.

References

1. A. Taha et al., “Energy Efficient Multipath Routing Protocol for Mobile Ad-Hoc Network Using the Fitness Function,” *IEEE Access*, vol. 5, 2017.
2. ns-3 Consortium, “ns-3 Network Simulator,” <https://www.nsnam.org/>.
3. Project scripts and source code in this repository (ns-3.39, FF-AODV modifications).